Standard Model of particle physics. The Higgs boson, if it were discovered, would, for reasons we'll get into shortly, finally confirm the leading theory explaining how fundamental particles were able to acquire mass in the early universe. And it would hopefully give us clues to the structure of physical law in regions beyond our current realm of exploration.

But that very prospect—probing the unknown reaches of reality—was enough to strike fear into the hearts of onlookers. No one had ever created collisions at these energies. No one knew how the laws of physics might shift and re-form themselves in such an environment.

Worst-case scenarios swirled around the internet. Perhaps the machine would open some kind of portal to another dimension, tearing apart the fabric of space itself. Perhaps it would create a tiny black hole that would grow, engulfing the entire planet. Perhaps it would create "strange matter"—a kind of composite material made of up-, down-, and strange-flavored quarks[fn1] that, some supposed, could lead to an ice-nine-style chain reaction,[fn2] converting all the matter it touched. But the physicists pressed on, apparently unconcerned. The LHC produced its first high-energy proton collisions in November 2009.

Given that life on this planet still exists, it's not too much of a spoiler to point out that none of the hypothesized existential disasters happened. (If you're still worried, there is a live-update website: www.hasthelargehadroncolliderdestroyedtheworldyet.com.) But did we just get lucky? Was the experiment really warranted, given the potential risks?

Physicists are not always cautious people, but exploring "what if" scenarios is kind of our bread and butter, and a chance to think deeply about the real physics behind hypothetical possibilities for ultimate destruction is too hard to pass up.[fn3] In fact, in 2000, four physicists (including one who would later win a Nobel Prize) wrote a sixteen-page paper for *Reviews of Modern Physics* called "Review of Speculative 'Disaster Scenarios' at RHIC." RHIC was the Relativistic Heavy Ion Collider, a Brookhaven National Lab collider that predated the LHC and that was built to collide the nuclei of heavy elements such as gold at high energies. A pioneering experiment in its own right, it was also the subject of worries that it might create unforeseen consequences that could endanger the planet (or the universe), and the paper was written to fully explore, and hopefully dispel, those rumors.